exploratory data analysis

hands-on data science · prof. dr. nicolas meseth

1. where exploring fits
2. what is the question?
3. exploring the survey

an explorer draws the map while walking, and every blank spot on it is the next question.



image: ai-generated (gpt-image-2.5-flare)

part 1

where exploring fits

the workflow of a data analysis

the workflow of a data analysis



exploring sits in the middle, and it is a loop, not a step.

after Wickham, Çetinkaya-Rundel & Grolemund: R for Data Science, 2nd ed.

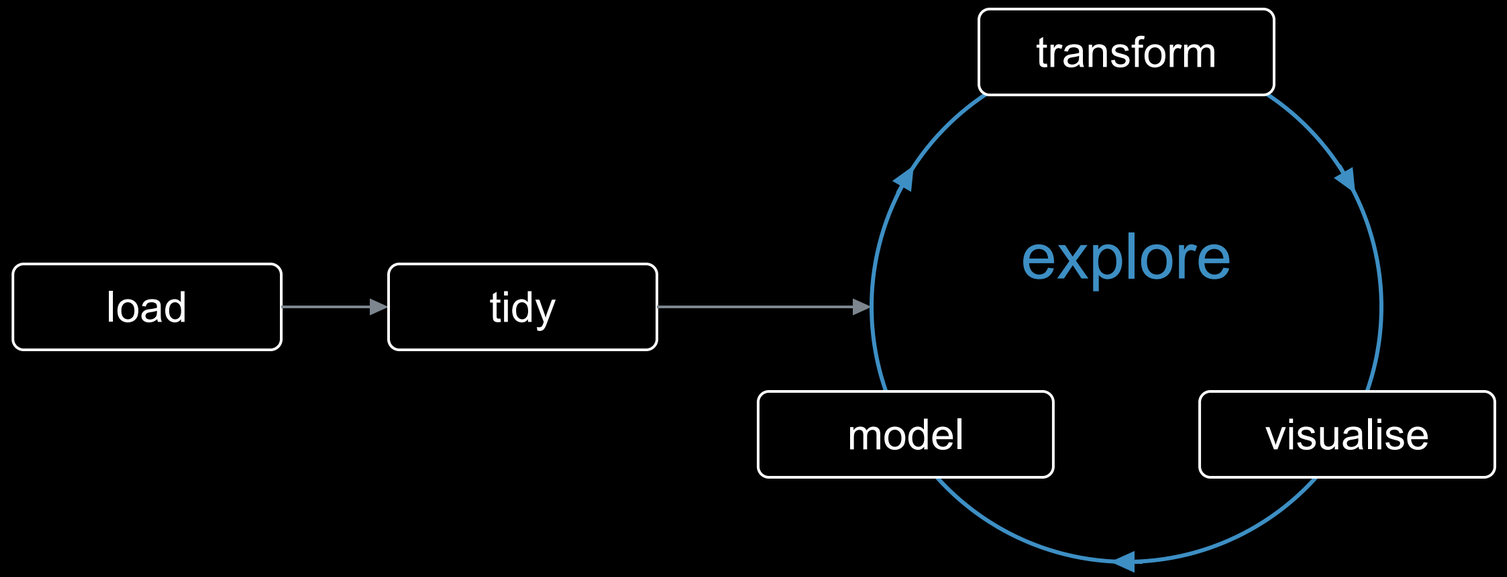
the workflow of a data analysis



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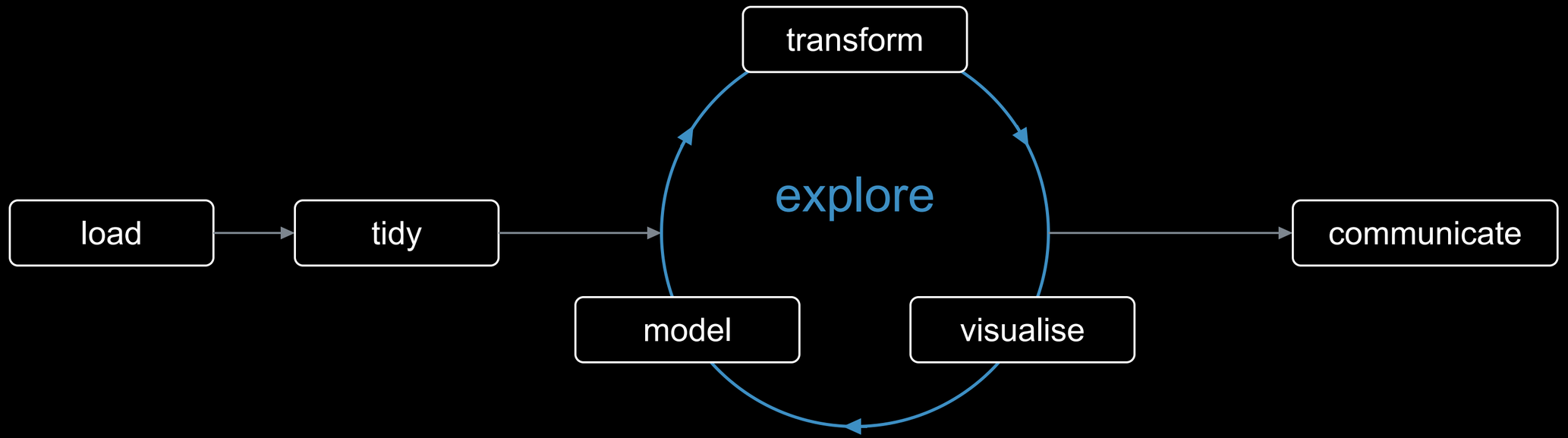
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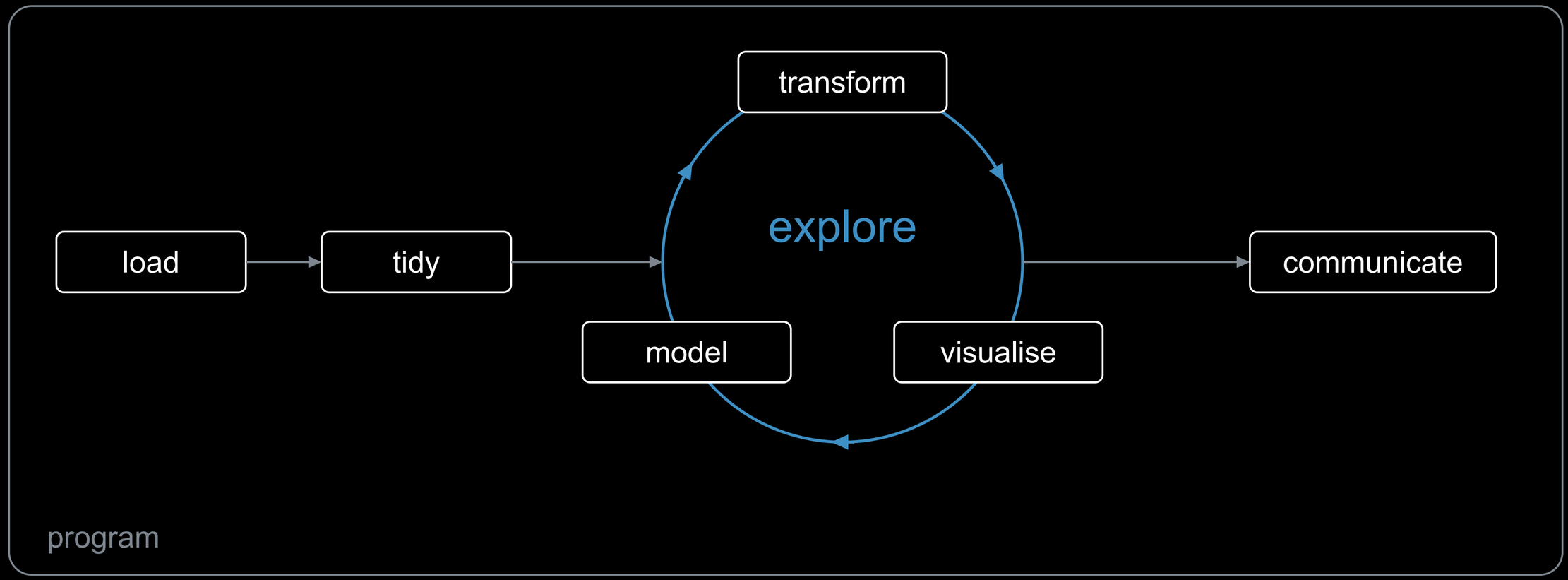
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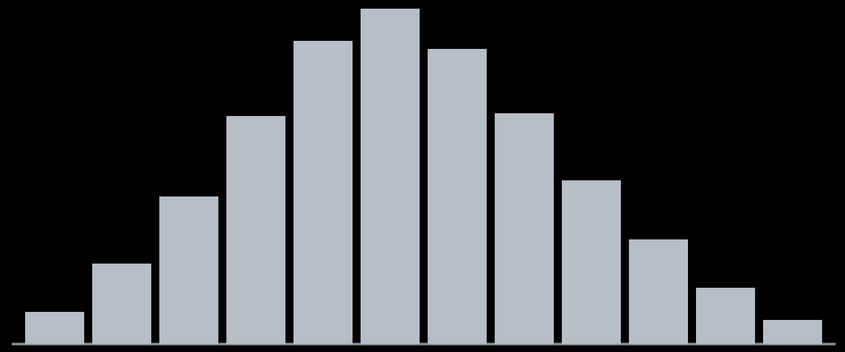


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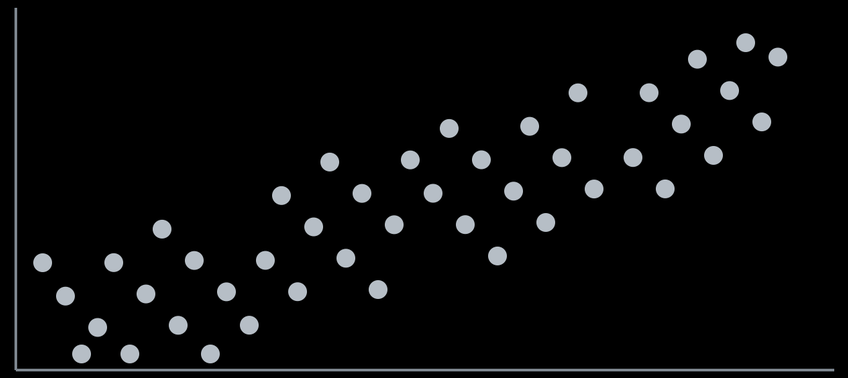
exploratory data analysis is a state of mind, not a fixed procedure.

R for Data Science: generate questions, search for answers by transforming and visualising, and use what you learn to refine the questions.



one column

what variation occurs
within one variable?



two columns

what covariation occurs
between variables?

almost every exploratory question asks how values vary, or how two variables vary together.

part 2

what is the question?

six types of data analysis

many analyses that are criticised later did not compute anything wrong, they answered a different question than the one people read into them.

Leek & Peng (2015): What is the question? Science 347, 1314–1315

six yes-or-no questions

did you summarise the data?

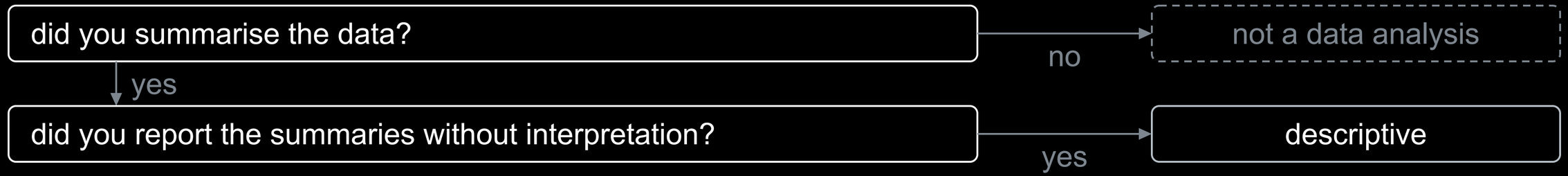
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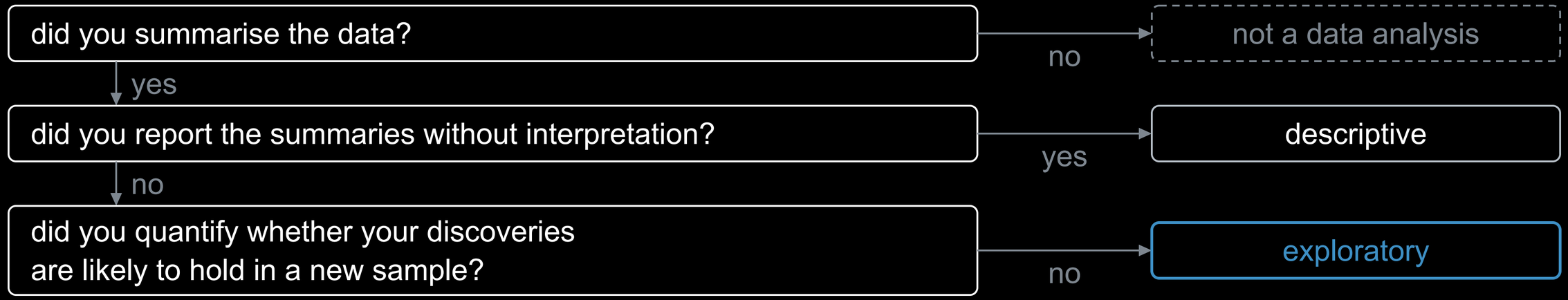
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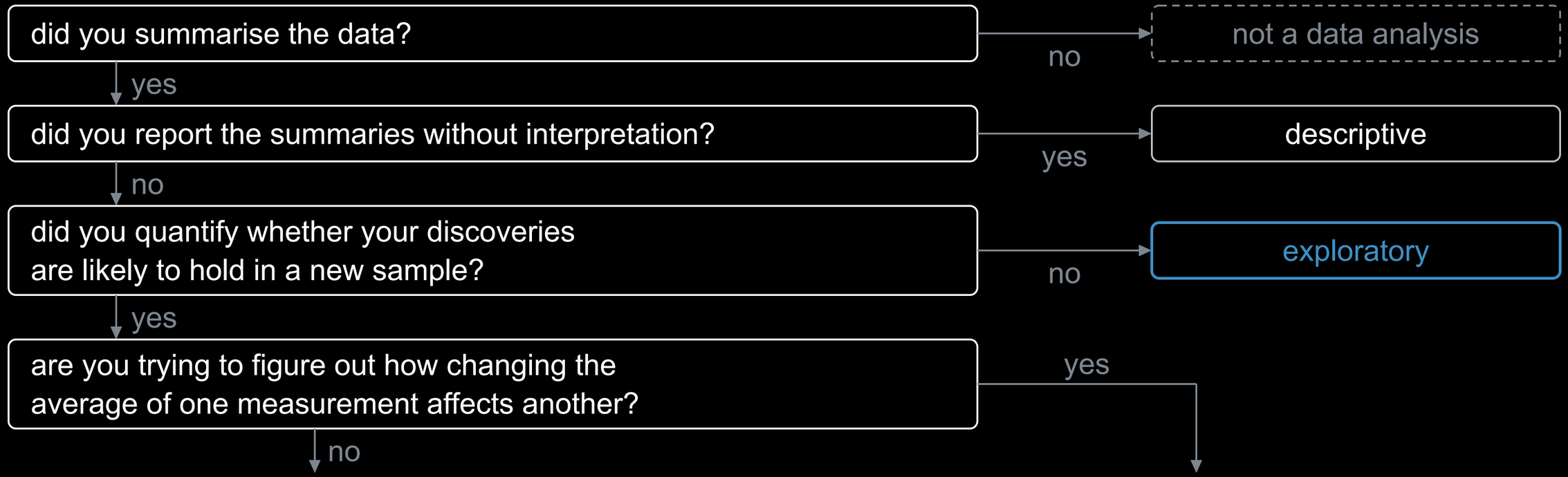
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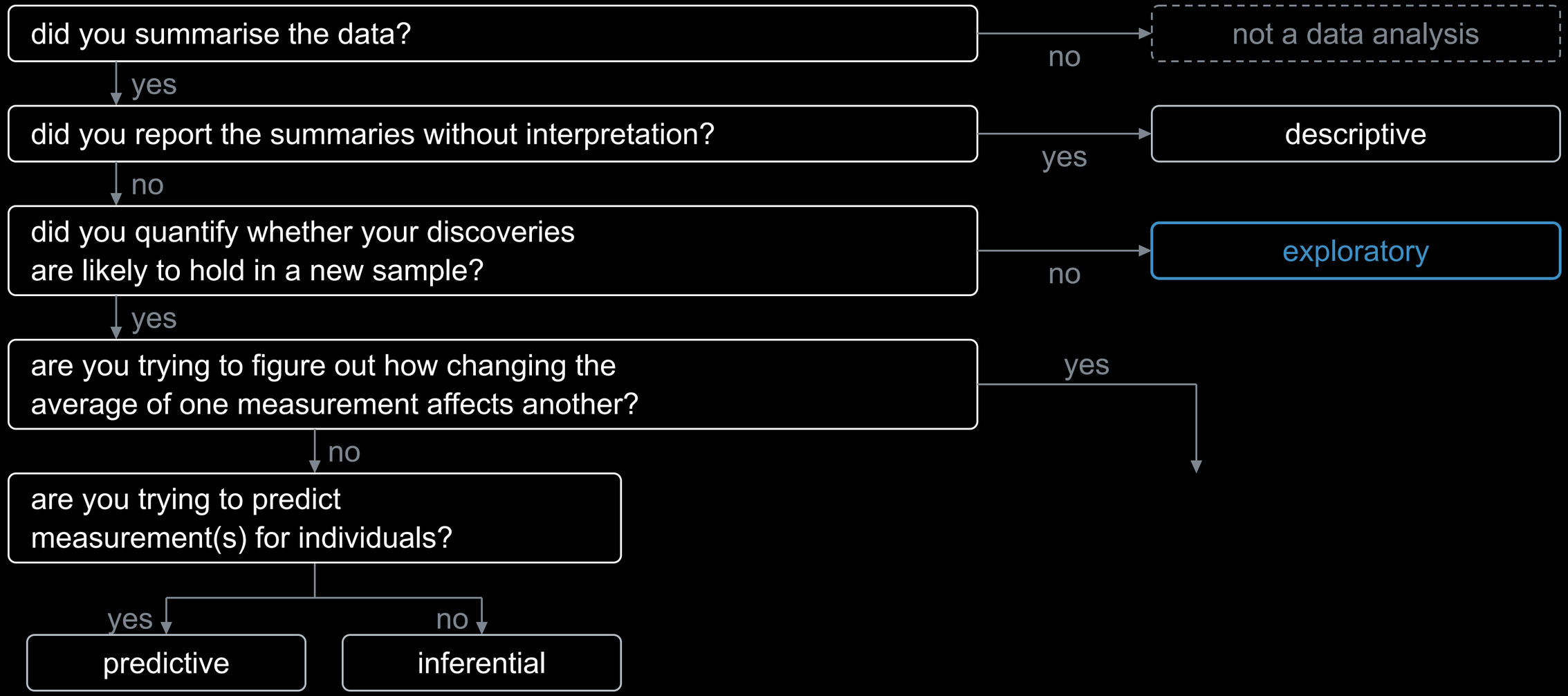
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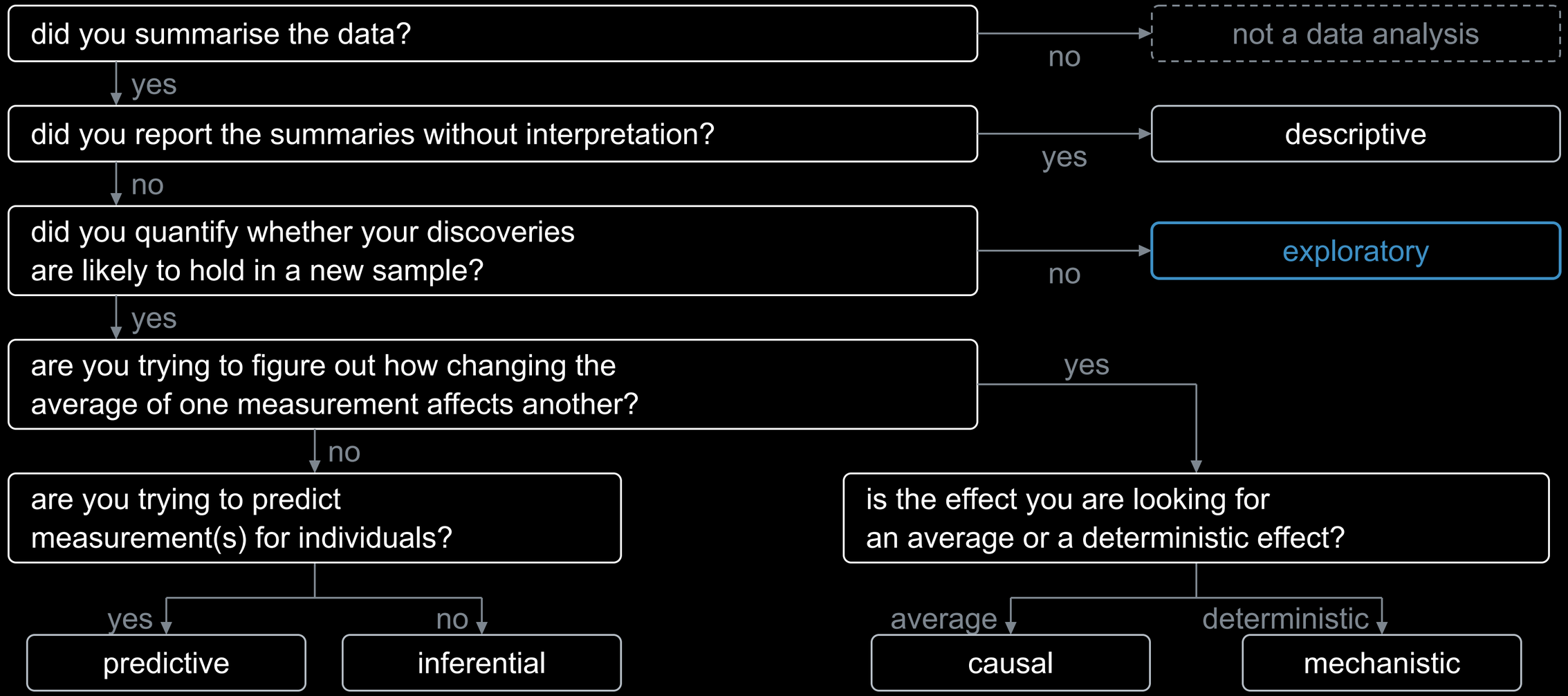
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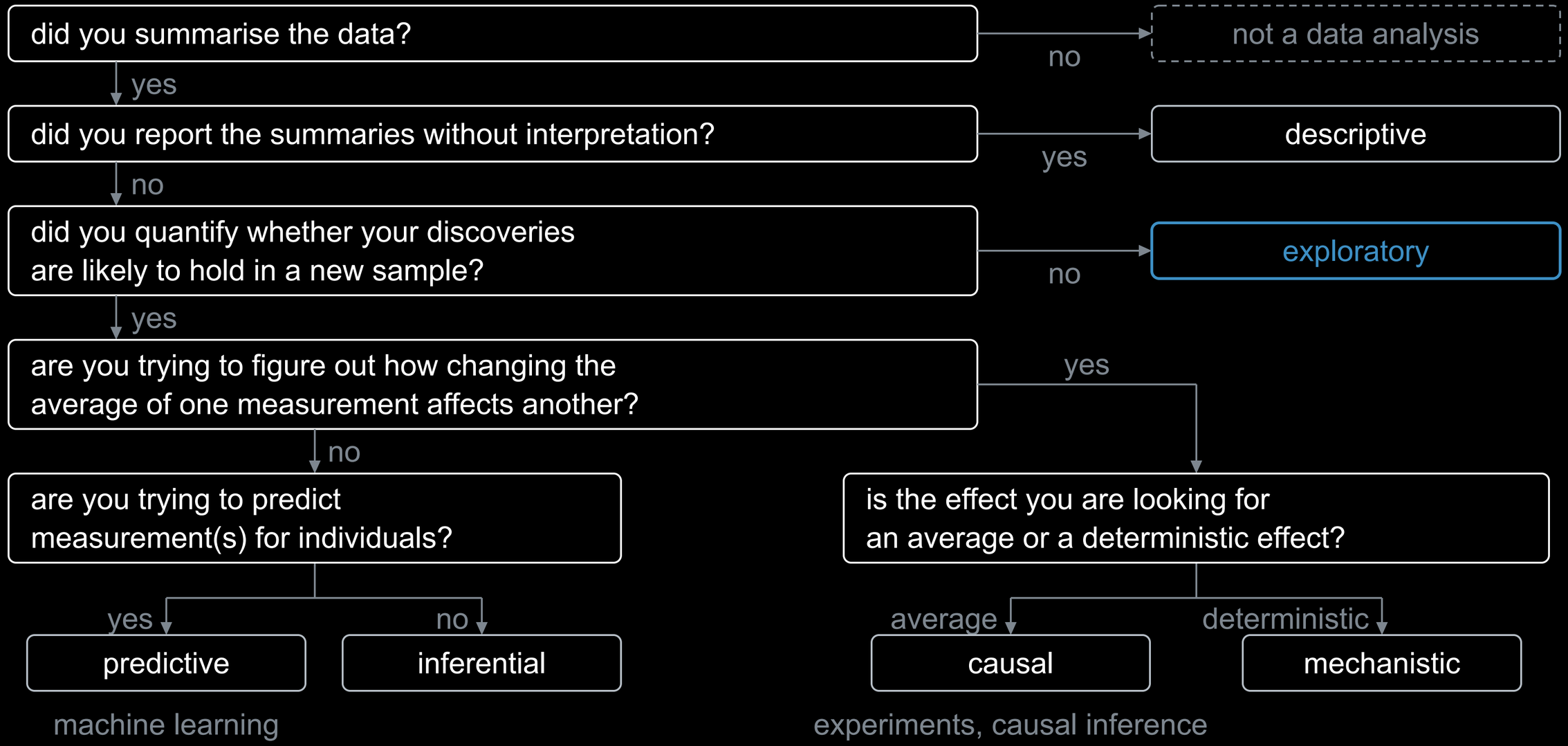
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six yes-or-no questions



after Leek & Peng (2015), Science 347, 1314–1315

six yes-or-no questions



after Leek & Peng (2015), Science 347, 1314–1315

six types, asked of one survey

type	what you do	a question for the milk survey
descriptive	count and summarise	how many drink milk every day?
exploratory	look for patterns	are the young more open to oat milk?
inferential	generalise beyond the sample	does that hold for all of Germany?
predictive	predict for individuals	can age and diet predict oat milk?
causal	change one thing, watch another	does a higher price change the choice?
mechanistic	understand each single case	how much energy heats a litre of milk?

which type of question is this: “what share of the respondents buys (almost) only organic food?”

descriptive

exploratory

inferential

causal

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the step people skip



holds for these 2,811 people

needs a test beyond the sample

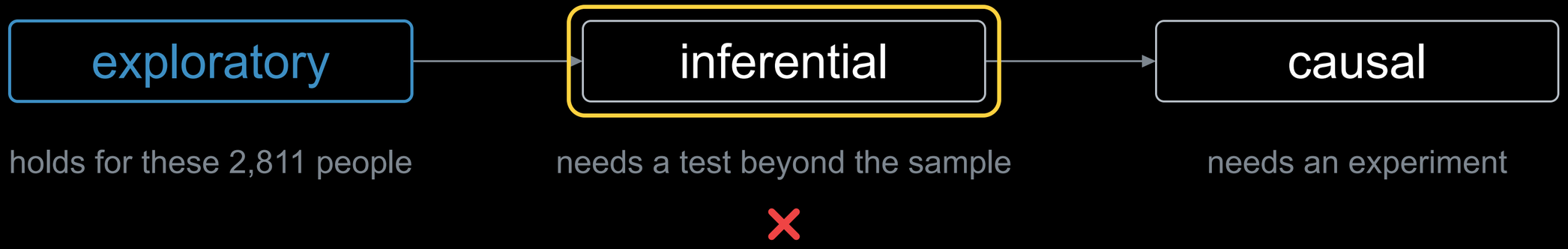
needs an experiment

the claim

in our survey, younger respondents are more open to oat milk.

the data stay the same, only the claim grows, and with it what you would have to show.

the step people skip

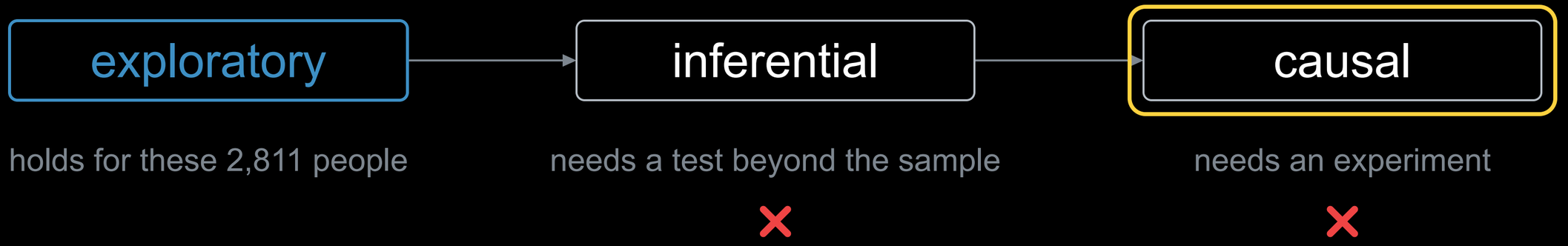


the claim

young people in Germany are more open to oat milk.

the data stay the same, only the claim grows, and with it what you would have to show.

the step people skip



the claim

being young makes people more open to oat milk.

the data stay the same, only the claim grows, and with it what you would have to show.

which type of claim is this: “vegetarians in Germany buy more organic food than others.”

descriptive

exploratory

inferential

causal

live.hsos.uber.space/q

which type of claim is this: “vegetarians in Germany buy more organic food than others.”

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exploratory

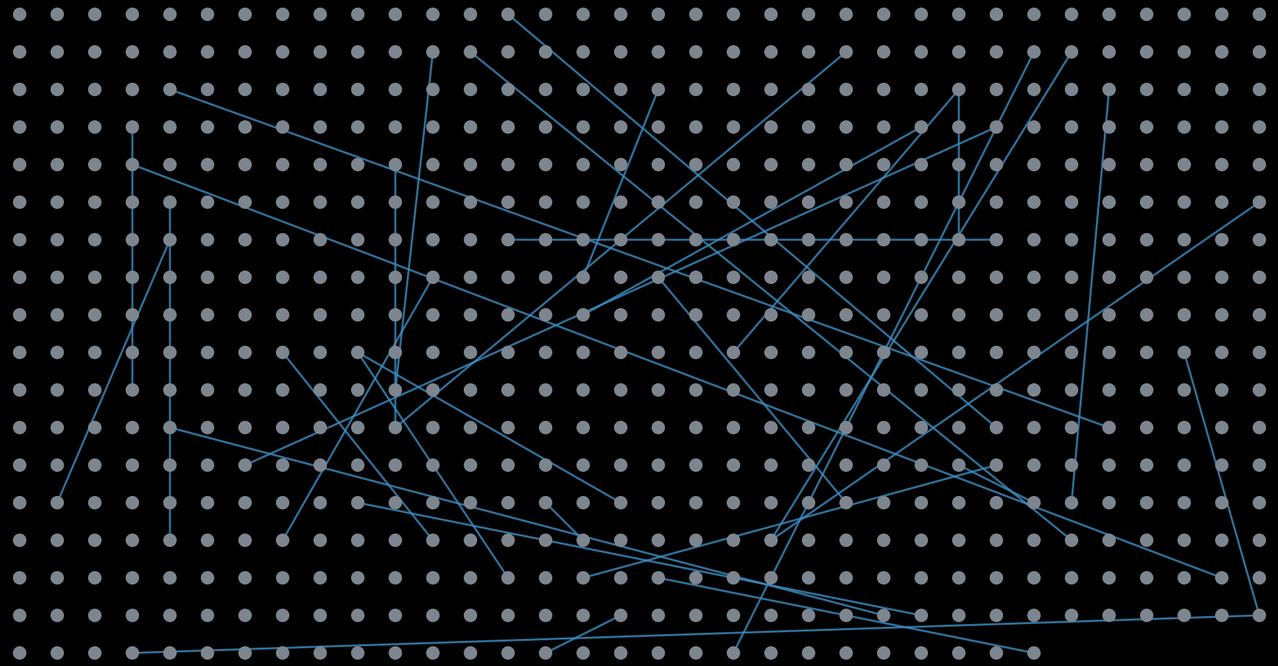
inferential

causal



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search long enough, and you will find something



606

columns

183,315

pairs of columns

about 7,000

pass a test at 5 % even though every column was shuffled

with 606 columns some pattern always appears, even if it lives only in these 2,811 people.

exploring finds questions worth asking, it does not answer them.

part 3

exploring the survey

variation, covariation, the unexpected

```
library(tidyverse)
survey <- read_csv("mds12_schoko_milch.csv")
dim(survey)
glimpse(survey)
```

```
[1] 2811  606
Rows: 2,811
Columns: 606
$ q001hheinkauf                 <dbl> 2, 1, 2, 2, 2, 2, 2, 2,…
$ q002geburt                    <dbl> 1970, 1990, 1963, 1989,…
$ q003land                      <dbl> 1, 7, 6, 13, 4, 2, 14, …
$ q004geschlecht                <dbl> 1, 1, 1, 2, 1, 1, 2, 1,…
```

before any question: how many rows, how many columns, which types, and what one row stands for.

readable names

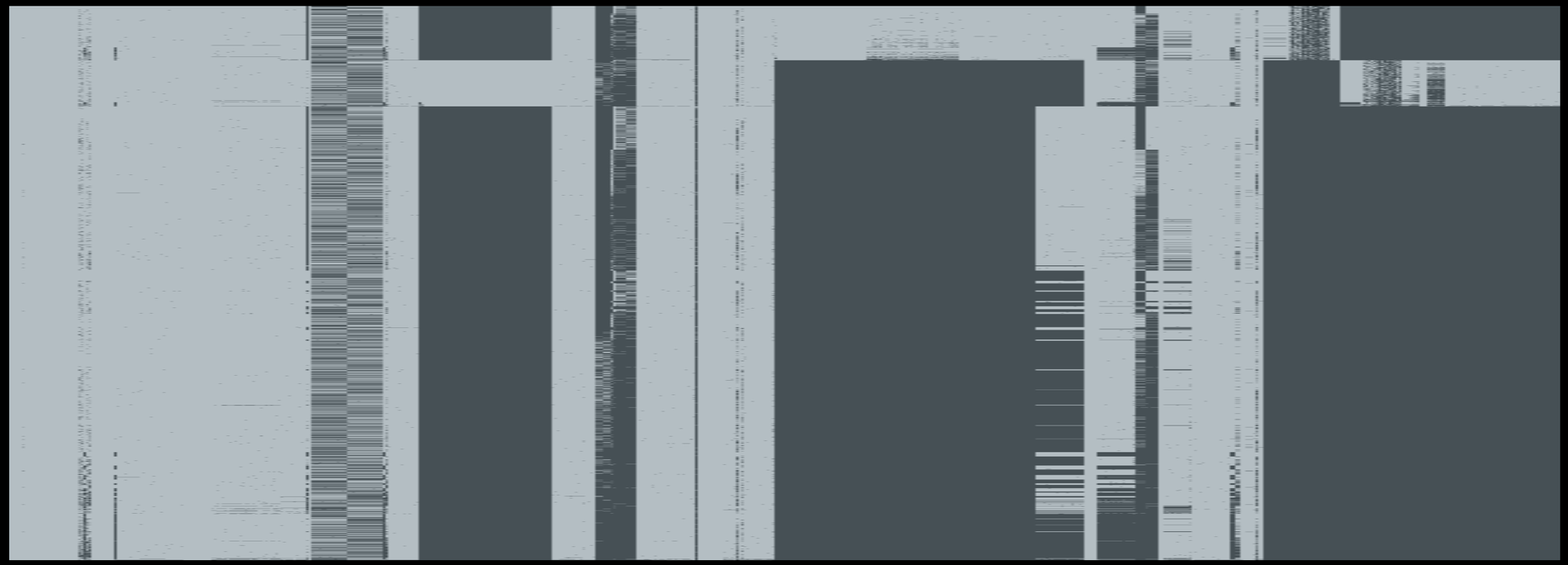
```
survey <- survey |>
  rename(
    age = Q002alter,
    organic_share = v032bioanteil,
    price_too_expensive = U017pvonwzuteuer2,
    no_oat_milk = m073haferpro1_nohafer
  ) |>
  mutate(diet = factor(
    v041nofleisch,
    levels = 0:4,
    labels = c("omnivore", "flexitarian", "vegetarian",
               "vegan", "pescetarian")
  ))
```

the codebook says what each code means; give the columns you need names you can read.

half of the table is empty

606 columns

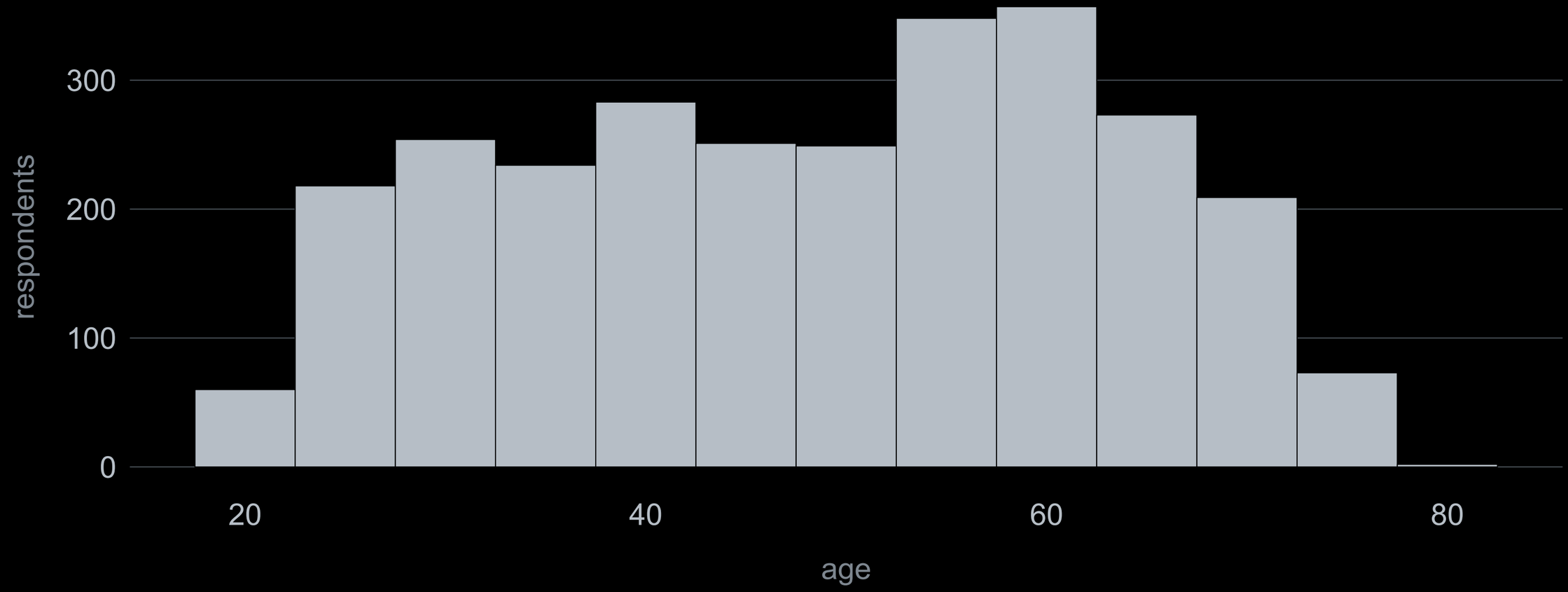
2,811 rows



answered empty

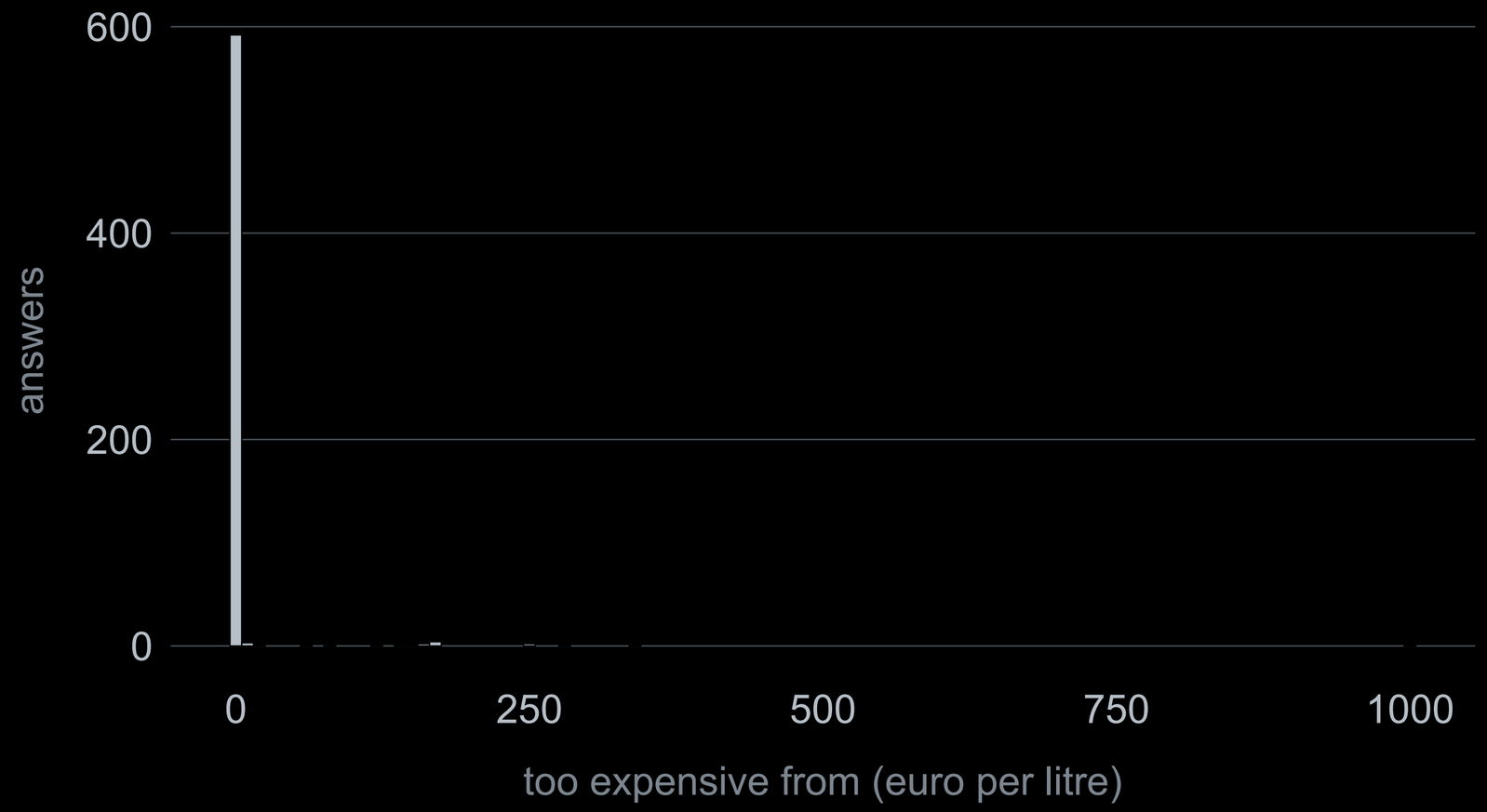
in a survey, missing usually means “not asked”, and dropping those rows changes who you are talking about.

variation: how old are the respondents?



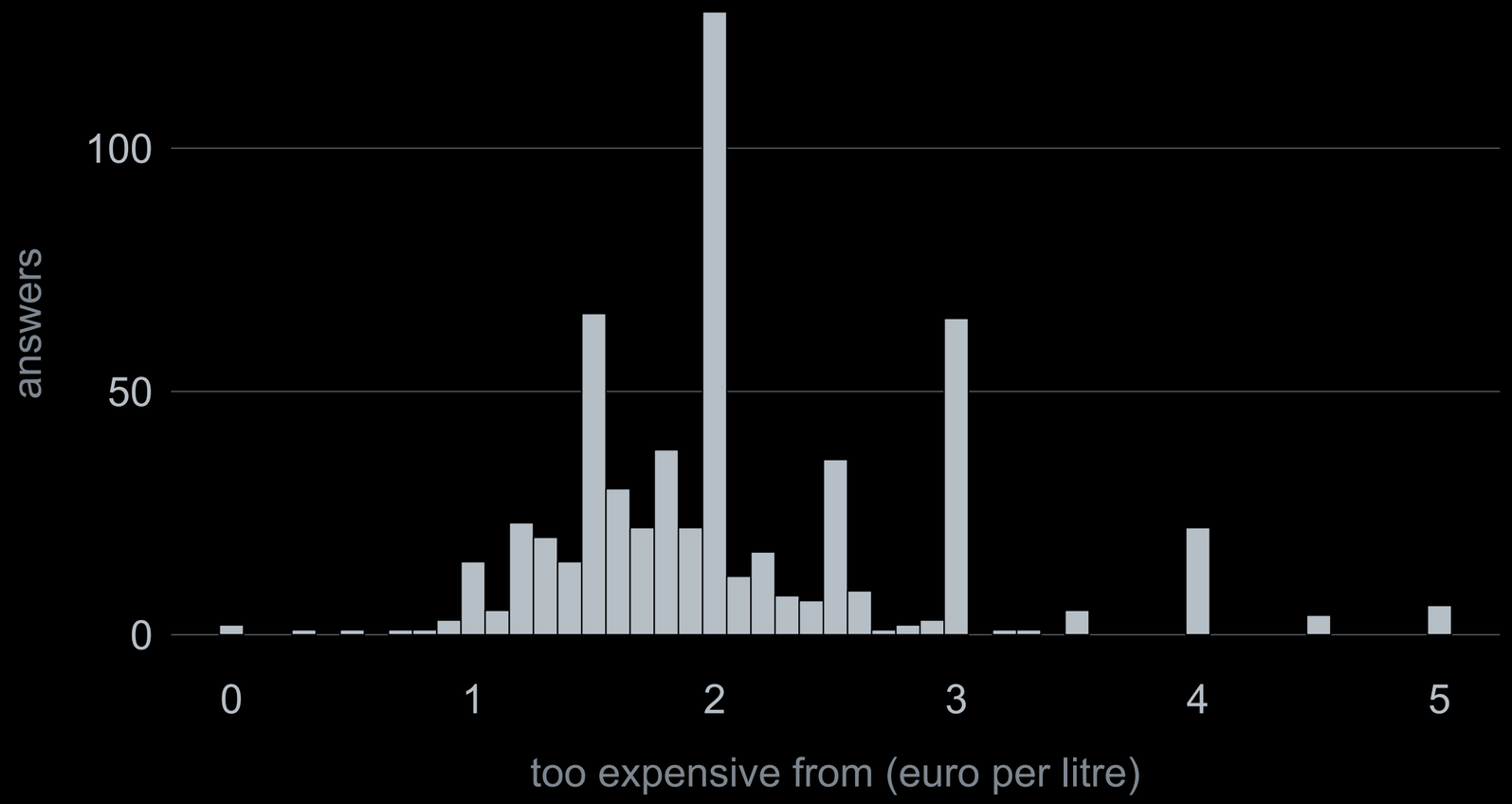
ages from 18 to 80 with the middle at 50: a quota sample, not a random one.

unusual values: when is milk too expensive?



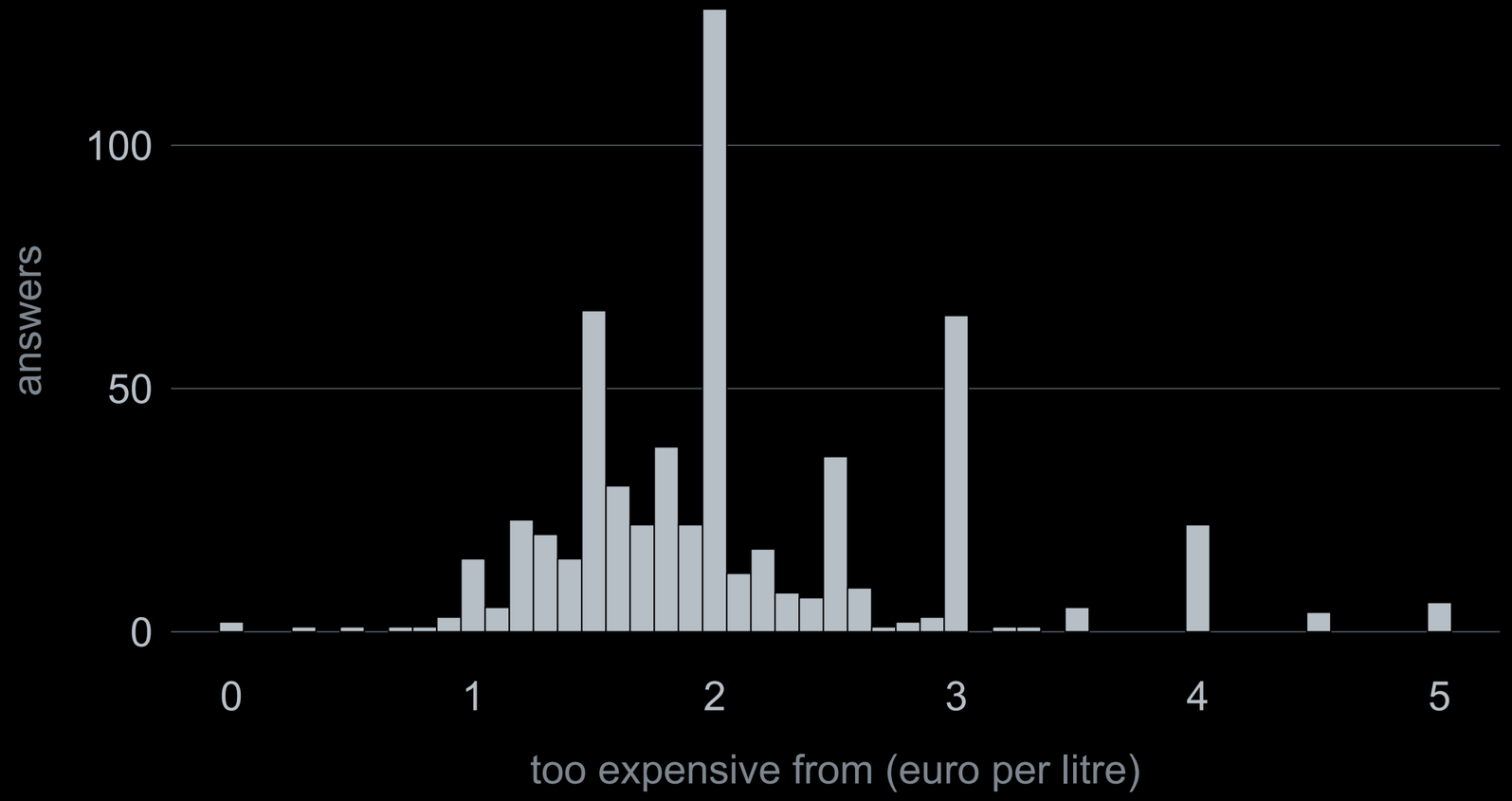
one strange value can hide a whole distribution, so look at the extremes before you look at the average.

unusual values: when is milk too expensive?



one strange value can hide a whole distribution, so look at the extremes before you look at the average.

unusual values: when is milk too expensive?



16 answers above 20

65	80	120	140
150	159	160	167
170	170	170	250
250	280	344	999

one strange value can hide a whole distribution, so look at the extremes before you look at the average.

deal with them, and say so

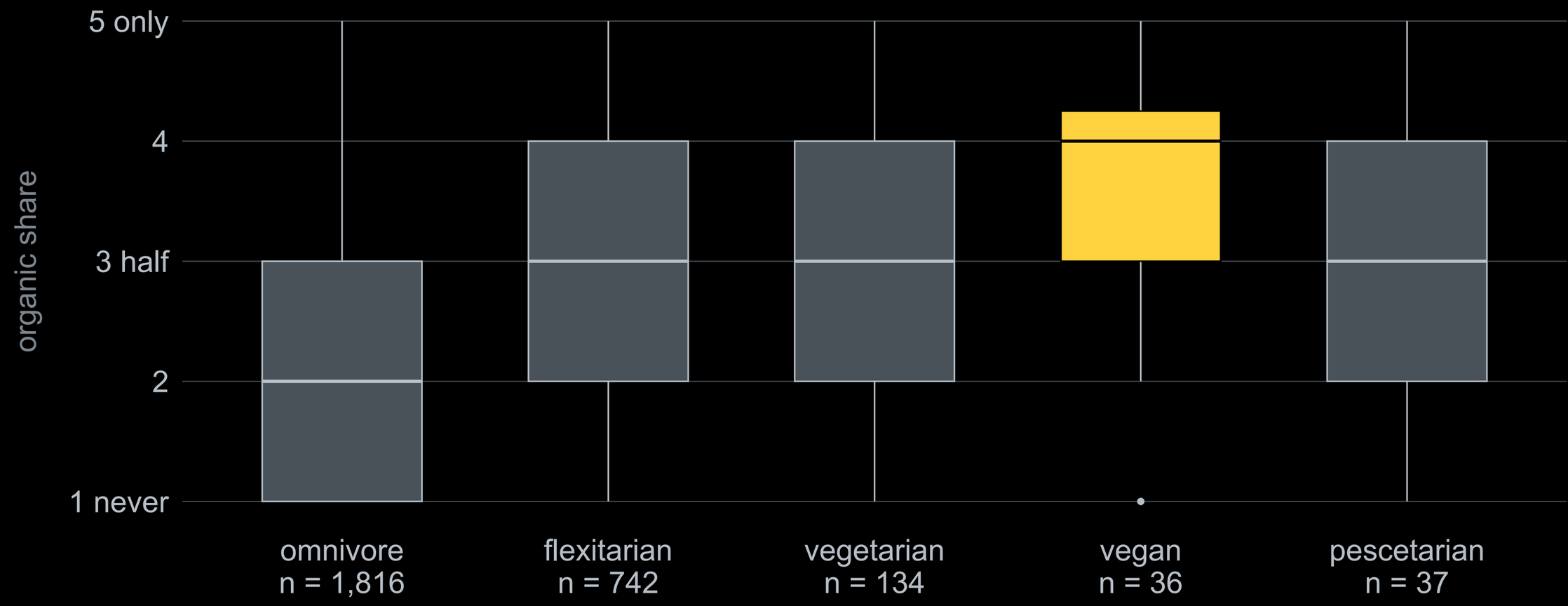
```
survey <- survey |>
  mutate(price_too_expensive = if_else(
    price_too_expensive > 20, NA, price_too_expensive
  ))
```

R for Data Science: replace unusual values with NA rather than dropping the rows, and report what you did.

	all 612 answers	16 answers above 20 set to NA
`mean()`	8.11 €	2.17 €
`median()`	1.99 €	1.99 €

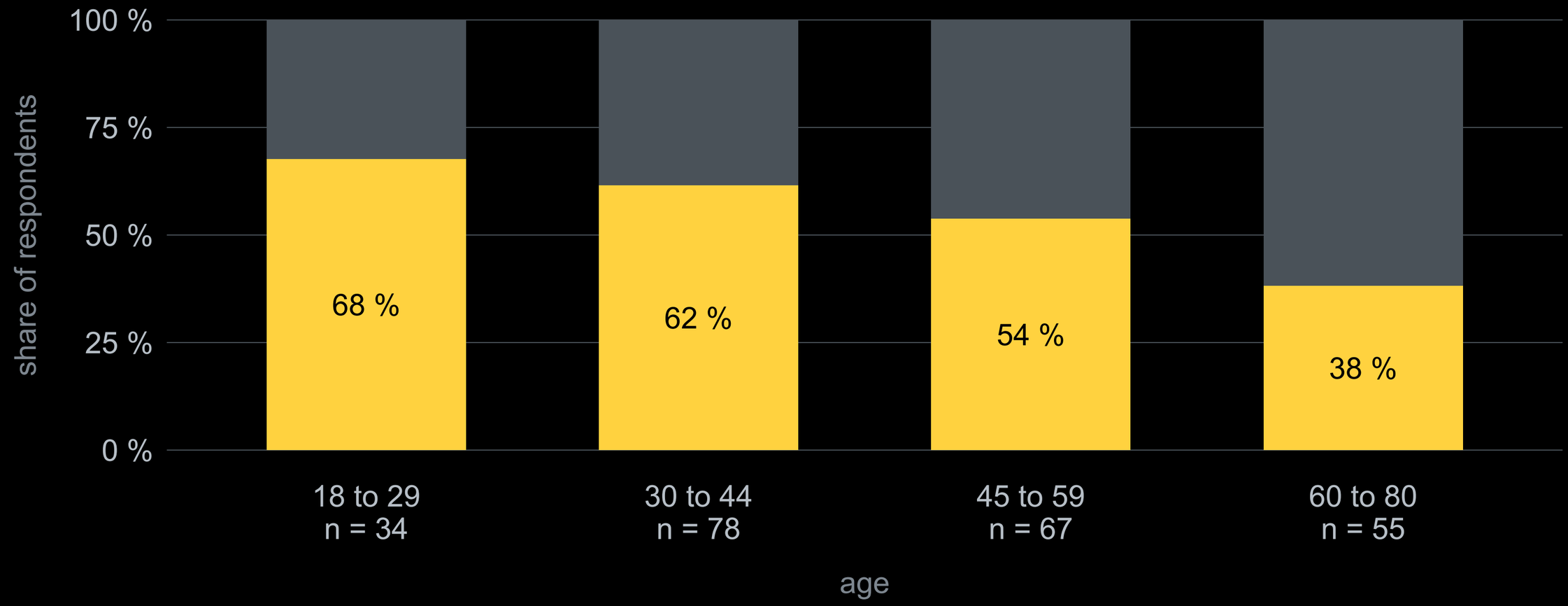
the median barely notices a typing error, while the mean is dragged along by it.

covariation: organic food and diet



the groups differ, and the smallest group differs most, which is exactly where chance plays the biggest part.

covariation: oat milk and age



a pattern in 234 people is a question for the next study, not yet an answer.

chance decided which price for alpro oat drink each respondent saw. which type of question can this answer?

descriptive

exploratory

predictive

causal

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from exploring to the next question

younger respondents are more open to oat milk

does it hold for all of Germany?

an inferential question

vegans and vegetarians buy more organic food

is it the diet, or the income behind it?

a causal question

16 price answers lie above 20 €

how should the questionnaire ask?

a question for the design

the result of an exploration is a list of better questions.

exploring finds questions worth asking, it does not answer them.

to read on: Leek & Peng (2015), “What is the question?”, and R for Data Science, chapter “exploratory data analysis”